

Project: Smart Garage Door System

Aim: To automatically operate a garage door using an ultrasonic sensor and DC motor controlled by an Arduino, enhancing safety, efficiency, and automation.

Objectives:

1. Circuit Design:
 - Develop a Tinkercad circuit that integrates an ultrasonic sensor, DC motor, Arduino Uno, MOSFET, and external power supply.
 - Ensure proper pin configuration and secure power management with protective elements (diode).
2. Proximity Detection Implementation:
 - Utilize the ultrasonic sensor to calculate object distance.
 - Trigger motor operation when an object is detected within a specified range (< 329 cm).
3. Motor Activation Logic:
 - Program Arduino to turn ON the motor for 4 seconds when an object (vehicle) is in range.
 - Ensure the door doesn't activate unnecessarily when no object is near.
4. Debugging and Feedback:
 - Use the Serial Monitor to log real-time distance data for troubleshooting.
5. Testing and Verification:
 - Simulate in Tinkercad by manually adjusting object distances.
 - Observe motor activity based on the proximity threshold.
6. Enhancements and Future Work:
 - Add limit switches to detect full open/close position.
 - Integrate LDR sensor to restrict motor operation in daylight.
 - Expand to Wi-Fi-based control using ESP8266 or Bluetooth modules.
 - Introduce smartphone notification or app-based override.
7. Project Presentation:
 - Demonstrate real-time simulation with visible serial output and motor response.
 - Explain hardware wiring and program logic clearly.

Procedures for Smart Garage System in Tinkercad :

1. Setup Tinkercad Account:
 - Visit [Tinkercad.com](https://www.tinkercad.com)
 - Create or log into an account.
 - Navigate to "Circuits" and click "Create New Circuit."
2. Gather and Add Components:
 - Arduino Uno R3
 - PING))) Ultrasonic Sensor

- DC Motor (with MOSFET switch circuit)
 - 10k Resistor
 - N-channel MOSFET
 - 1N4007 Flyback Diode
 - 9V Battery
 - Breadboard, Jumper Wires
3. Build the Circuit:
- Ultrasonic Sensor Wiring:
- VCC → 5V (Arduino)
 - GND → GND
 - SIG → Digital Pin 2
- Motor Control Wiring:
- Motor → MOSFET Drain
 - MOSFET Source → GND
 - Gate → Digital Pin 13 via 10k resistor
 - External +9V → Motor (+)
 - Diode across motor terminals (cathode to +ve)
4. Set Pin Modes in Code:
- Pin 2 as Input for distance detection.
 - Pin 13 as Output for motor activation.
5. Logic Flow:
- Sensor sends pulse.
 - Arduino measures echo time.
 - Converts to distance.
 - If distance < 329 cm, motor runs 4 seconds.
 - Else, motor stays OFF.
6. Simulation:
- Start simulation in Tinkercad.
 - Adjust simulated object proximity.
 - Observe motor on/off actions.
 - Read serial monitor logs.
7. Debugging Tips:
- Check ultrasonic sensor wiring and pin mapping.
 - Verify motor power source and MOSFET orientation.
 - Use Serial.print() for distance values.

Arduino Code :

```
#include <Stepper.h>
```

```
const int pingPin = 2;  
const int motorPin = 13;
```

```
void setup() {
  // initialize serial communication:
  Serial.begin(9600);
  pinMode(motorPin, OUTPUT);
}

void loop() {
  // establish variables for duration of the ping,
  // and the distance result in inches and centimeters:
  int duration, cm;

  // The Sonar is triggered by a HIGH pulse of 2 or more microseconds.
  // Give a short LOW pulse beforehand to ensure a clean HIGH pulse:
  pinMode(pingPin, OUTPUT);
  digitalWrite(pingPin, LOW);
  delayMicroseconds(2);
  digitalWrite(pingPin, HIGH);
  delayMicroseconds(5);
  digitalWrite(pingPin, LOW);

  // The same pin is used to read the signal from the Sonar: a HIGH
  // pulse whose duration is the time (in microseconds) from the sending
  // of the ping to the reception of its echo off of an object.
  pinMode(pingPin, INPUT);
  duration = pulseIn(pingPin, HIGH);

  // convert the time into a distance
  cm = microsecondsToCentimeters(duration);

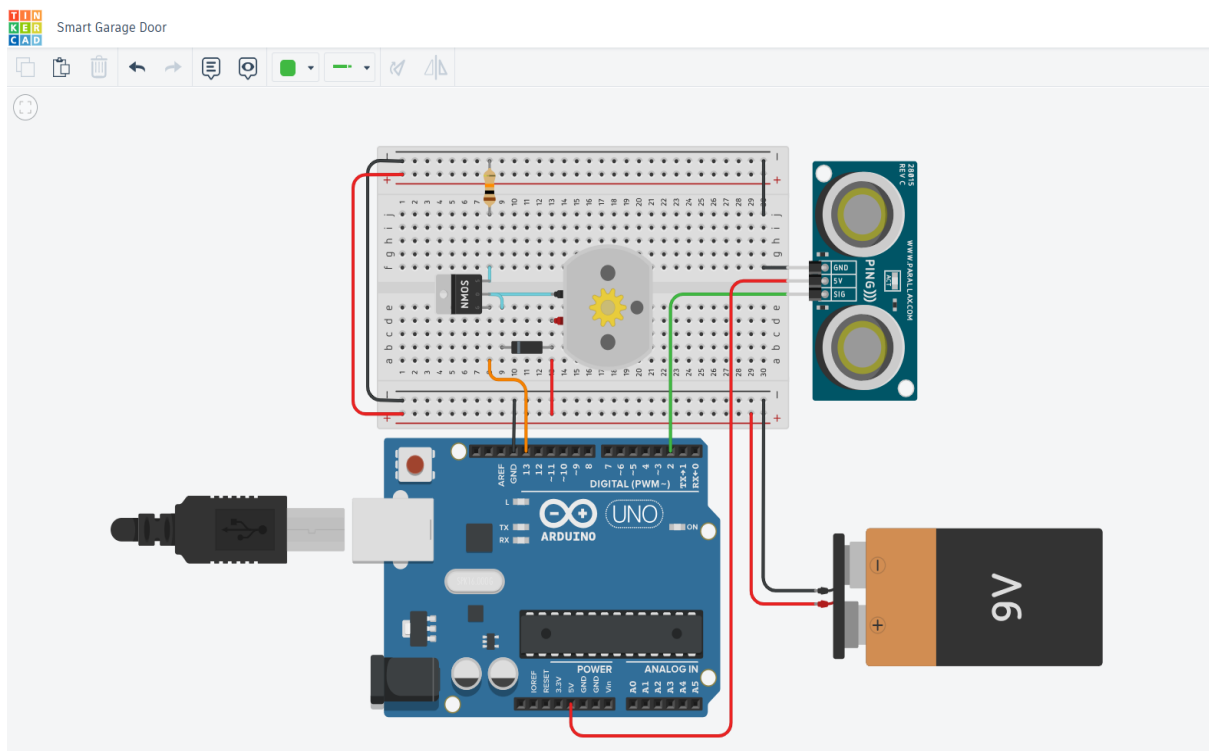
  // Print the distance
  Serial.print("Distance: ");
  Serial.print(cm);
  Serial.print("cm");
  Serial.println();

  // Turn on the motor when the object is in range
  if(cm < 329) {
    digitalWrite(motorPin, HIGH);
    delay(4000); // The time it takes for the garage door to reach the top or bottom
  }
  else {
    digitalWrite(motorPin, HIGH);
  }
  digitalWrite(motorPin, LOW);
  delay(1000);
}

long microsecondsToCentimeters(long microseconds) {
  // The speed of sound is 340 m/s or 29 microseconds per centimeter.
```

```
// The ping travels out and back, so to find the distance of the
// object we take half of the distance travelled.
return microseconds / 29 / 2;
}
```

Hardware & Simulation Result:



Expected Behavior:

Scenario	Motor Output	Serial Output
No object nearby	OFF	Distance: > 329 cm
Object detected close ON (4s)		Distance: < 329 cm

Observations:

- Sensor accurately detects object distance.
- Arduino controls motor based on proximity.
- Serial output confirms system logic in real-time.
- Delay avoids repeated false triggers.

Advantages:

1. Automatic garage door control
2. Cost-effective and simple
3. Hands-free operation
4. Simulation-friendly via Tinkercad
5. Easily expandable

Disadvantages:

1. No real-time feedback (no display)
2. Lacks position feedback for motor
3. No app/manual override
4. Detection limit based on fixed range

Conclusion: This Smart Garage Door system demonstrates effective use of sensor-based automation using Arduino. It can serve as the base for advanced IoT-enabled, app-controlled home automation systems. The design is energy efficient, responsive, and beginner-friendly.
